Type II conditionals in writings by beginner and advanced English learners: A collostruction-based analysis

Daniela Verratti Souto

1 Introduction

1.1 Learner corpora in Second langauge acquisition

Collostructional analysis: Stefanowitsch and Gries (2003)

> "For studies that aim to contribute to SLA scholarship by providing insights into the progress of L2 acquisition, it will be important to analyze and compare data from learners at different proficiency levels, ideally following a longitudinal design and using corpora that capture data collected from the same learners at different points in time."

(Römer-Barron, 2026)

conditionals more non-native like (Gries and Wulff, 2021)

1.2 Formulaicity in learner language

beginThese data showed that multimorphemic sequences that go well beyond learners' grammatical competence are very common in early L2 production. Notwithstanding that these sequences contain such forms as finite verbs, wh-questions and clitics, Myles denied this as evidence for functional projections from the start of L2 acquisition because these properties are not present outside chunks initially. Analyses of inflected verb forms suggested that early productions containing them were formulaic chunk (Ellis, 2012)

1.3 Collostructional analysis

The present study is pseudo-longitudinal in nature and intends to observe, through a collostructional lense, aspects of the development of the second conditional in L2 English learners.

RQ1. How are CIIs dispersed across learner writing tasks and proficiency level in our analyzed data?

RQ2. Do learners at higher proficiency levels favor different verbs in the protasis of CIIs than beginner learners?

RQ3. Does the verbal slot in the protasis of CIIs provide evidence for a transition from more formulaic to more productive use of the construction?

2 Methods

2.1 Data: The EFCAMDAT

The EF-Cambridge Open Language Database (EFCAMDAT), first presented by Geertzen et al. (2013), is a large-scale learner corpus containing writings from the online English school *Englishtown*. In their courses, each level corresponds to eight pedagogical units, after each of which students receive a prompt for a writing task that they must complete in order to advance. The corpus comprises responses to these tasks in their original and corrected forms, along with metainformation about the text and the author. An advantage of the EFCAMDAT, beyond being generally available for researchers, is the fact that *Englishtown* levels are aligned with the Common European Framework of Reference (CEFR), making a broad range of descriptive analyses possible at globally understood proficency levels.[1]

This study used the version of EFCAMDAT cleaned and post-processed by Öksüz et al. (2025), which contains 620,206 learner writings. However, it was unlikely for CIIs to be elicited by most prompts; therefore, after qualitatively inspecting the task prompts, I decided on a subset of 20 tasks whose responses were most likely to contain the construction. The selected units and corresponding writing topics are outlined in Table 3 in Appendix A, totaling 33,496 student writings and 4,375,055 words stemming from CEFR levels A2 to C1.

Finally, to ensure sufficiently large samples of both learner texts and CII instances, the writings were grouped into two proficiency bands: learners at CEFR levels A2 and B1 ("lower") and learners at levels B2 and C1 ("upper").

2.2 Extracting type II conditionals

All corrected learner writings in the subcorpus were automatically parsed using Stanza (Qi et al., 2020). Using the corrected texts instead of the original ones is justified by the fact that accuracy does not factor into this study: the goal is not to assess grammatical proficiency, but rather to explore differences in lexical preferences across levels. This allows us to use the corrected versions of the texts, ensuring more robust output from NLP tools.[2]

[1] The availability of large-scale, CEFR-aligned corpora is more limited than one might expect. The most well-known English learner corpus is the Cambridge Learner Corpus (CLC), whose access is highly restricted.

[2] However, given previous robustness studies of NLP tools on L2 English (Geertzen et al., 2013), similar results would be expected when running the analysis on the original learner texts.

The parsed documents were scanned for CII using both Universal Dependency (UD) relations and part-of-speech (POS) information. Only sentences containing the word "if" as a conditional marker (headed by an `advcl` UD tag, as in Figure 1) were taken into account, excluding interrogative markers headed by a `ccomp` UD tag, as in Figure 2. Then, a Python script ensures that the verb in the if-clause is in the past tense and the main clause contains a modal verb, thus filtering out other types of conditionals. For this study, I excluded sentences that expressed their conditional value through means other than the grammatical marker "if" (e.g. subject-auxiliary inversion) and sentences containing a modal verb in the protasis (e.g. *If I could go anywhere in the world, ...*).

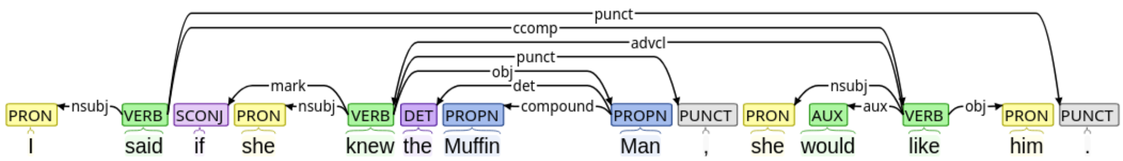

Figure 1: UD parse of a sentence with a conditional *if*

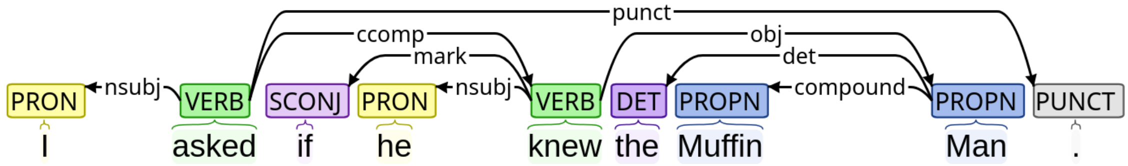

Figure 2: UD parse of a sentence with a non-conditional *if*

Finally, conditionals that fit all requirements were compiled into a CSV file with the full sentence, the lemma filling the verbal slot in the protasis,[3] and relevant metadata about the writing prompt and response, such as the *Englishtown* level the task stemmed from and the topic ID.

2.3 Calculating dispersion across tasks and levels

As a task-based corpus, the texts from EFCAMDAT are subject to task effects (Alexopoulou et al., 2017). That is, the educational environment in which the texts are embedded shapes the grammar and lexis found in the corpus. When drawing conclusions from this kind of corpora, it is possible to run into issues if one assumes all texts to be representative of the general production of learners at their corresponding proficiency, regardless of the task. To observe the effect of task on the frequency of CIIs, I applied Gries's (2008) DP (deviation of proportions) measure and its normalized version DP_{norm} for corpus parts of unequal size as presented in the correction by Lijffijt and Gries (2012). The same process was also applied using the binary proficiency levels "upper" and "lower" as the corpus parts to ensure that the Distinctive Collexeme Analysis was carried out in comparable parts of the analyzed subcorpus.

[3] In the case of constructions with auxiliary verbs, I chose the lemma of the lexical verb as the slot filler in order to avoid artificially inflating the auxiliary's frequency and more accurately reflect lexical variability.

2.4 Comparing tendencies across two L2 English proficiency levels

To investigate the differences in verb preferences in the if-clause of CIIs, I used Distinctive Collexeme Analysis (DCA), an extension collostructional analysis which compares a collexeme's degree of attraction to one construction with its attraction to an alternative, similar construction (Gries and Stefanowitsch, 2004). For the purposes of this study, however, instead of comparing two analogous but distinct constructions, we compare collexemes' association strength to the same constructional slot across two different proficiency levels.

In recent years, Gries has devised a more computationally-efficient way of calculating association strengths in collostructional analysis by replacing contingency table-based computations with the residuals of the χ^2-test, achieving extremely high correlations with the results of the traditional technique (Gries, 2023). Given its efficiency and relative simplicity, the present study adopts this updated approach.

To quantify the lexical diversity of verbs filling the slot under discussion, I opted for an entropy-based approach. Information-theoretic entropy (H) measures the uncertainty in a probability distribution: the more evenly distributed, the higher the entropy. When normalized to a 0-1 scale (H_{rel}), entropy values become comparable across distributions with different numbers of categories (Gries and Ellis, 2015). Entropy-based measures have been used in SLA corpus research to investigate L2 development (for example, see Sun and Wang (2021) for an analysis of entropy and complexity done on the EFCAMDAT) and was thus deemed an appropriate measure for the purpose of comparing the breadth and distribution of verb lemmas in the relevant position.

3 Results

A total of 842 instances of CIIs were found in the parsed subcorpus. The current section outlines the results of my analysis.

To examine the evenness of occurrence of CIIs across tasks and proficiency levels, I calculated dispersion indices (DP and DP_{norm}) using different definitions of corpus "parts". First, I partitioned the data into the 20 task topics. Second, the data was divided by proficiency level (lower vs. higher). Finally, I calculated topic dispersion separately within each proficiency group. Table 1 reports DP_{norm}, since raw DP values are nearly identical.

Overall, dispersion across task topics was relatively high, suggesting that different tasks elicited considerably different numbers of CIIs in their responses. Nevertheless, when stratifying the tasks by proficiency levels, higher-level tasks showed a more uneven distribution of this construction than lower-level ones. On the other hand, dispersion across levels was very low, revealing that both proficiency groups produced a very similar number of conditionals to what would be expected by their corresponding size.

Partitioning / subset condition	DP_{norm}
Topics across all data	0.559
Proficiency level as partitioning variable	0.022
Topic dispersion within proficiency groups	
Low proficiency (topics)	0.182
High proficiency (topics)	0.376

Table 1: Dispersion of type II conditionals. The last block shows topic dispersion calculated separately for low- and high-proficiency subsets.

Association strengths between the verbal collexemes in the if-clause of CIIs in writings by higher- and lower-proficiency students were contrasted by means of DCA using the residuals of a χ^2-test. Figure 3 illustrates the results of those verbs with at least five occurrences in the dataset. Positive residuals indicate attraction toward conditionals produced by upper-level learners, whereas negative residuals indicate that the verb was favored by lower-proficiency students.

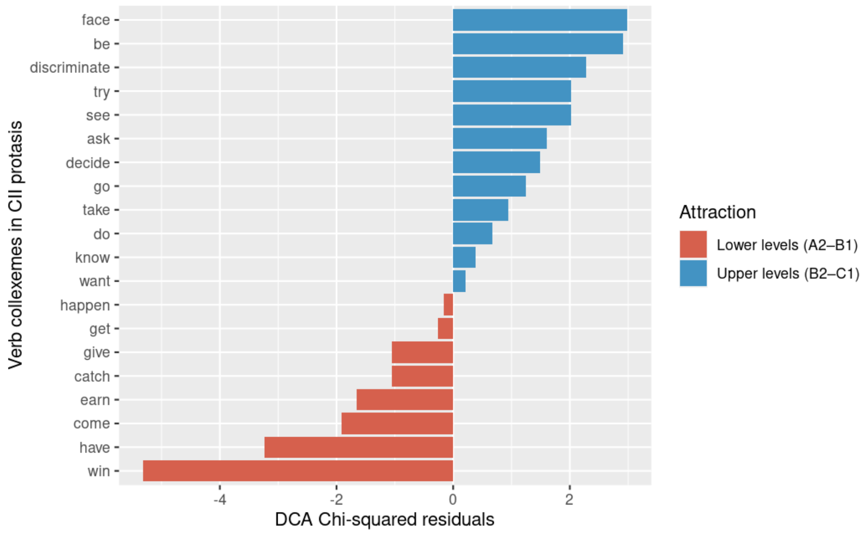

Figure 3: Residuals of the χ^2-test for DCA

The analysis reveals a clear attraction of a few collexemes to one or the other group of students. Advanced students disproportionately favored some higher-level words like "face" or "discriminate", as well as some high-frequency verbs like "be", "try" and "see". On the other hand, the verb "win" was notably overrepresented in texts written by beginners.

Normalized Shannon's Entropy (H_{rel}) was used to measure the diversity of verbs used by beginner and advanced students as a proxy for measuring a potential shift from formulaicity to productivity in CIIs. The results are summarized in Table 2 below.

Proficiency level	Entropy (H)	Normalized entropy (H_{rel})
Lower	2.391	0.628
Higher	2.713	0.639

Table 2: H and H_{rel} values for verb distributions across proficiency groups.

The raw entropy values for the advanced learner texts is slightly higher than for beginner writings. This difference becomes smaller, however, when the entropy values are normalized, suggesting that higher-proficiency students produce overall a larger number of unique verb types, though the evenness of the distribution at both levels is similar.

4 Discussion

Observing the high dispersion of CIIs across task topics reveals clear task effects: the number of conditionals elicited by each task is indeed not proportional to what would be expected simply from the number of words in their responses. The most likely explanation for this is that some tasks simply require more reasoning about imaginary or unlikely events in the present or future than others. Similar conclusions have been reached regarding the EFCAMDAT not in terms of a specific construction, but with a general focus on different measures of linguistic complexity (Alexopoulou et al., 2017). An alternative reason for an uneven distribution of CII across tasks is the explicit introduction of the construction within specific units, resulting in students being primed to produce a greater number of sentences containing this specific structure. These are not mutually exclusive, but rather, it is to be expected that learning material designers intentionally provide students with opportunities to use newly-introduced grammar means.

On the other hand, perhaps surprisingly, this study reports the opposite pattern regarding student level: CIIs appear to be produced in similar proportions by students in lower and higher levels of proficiency. It is very likely that this observation is due to the second conditional being acquired at an intermediate level like B1, as suggested by the English Grammar Profile (O'Keeffe and Mark, 2017), and this is where most of the tasks and condtionals from the lower-proficiency level in this study come from. Then, once this construction is available to the learners, its use is maintained in further stages of L2 development, since few alternative structures exist that fulfill the same communicative function.

However, despite the relatively stable overall frequencies observed across proficiency levels, subsequent analyses revealed important differences in the lexical composition of CIIs. The DCA results for higher-proficiency learners included verbs such as *face* (e.g., *face one's fears*) and *discriminate*, classified at the B2 and C1 levels, respectively, in the English Vocabulary Profile (Capel, 2010, 2012). At the same time, highly frequent verbs such as *try* and *see* were also among the most strongly associated verbs. Most interestingly, perhaps, copular *be* was the second most strongly attracted verb to writings by advanced learners. Conversely, in beginner texts, the most highly-attracted fillers were all common lexical verbs.[4].

To better understand these patterns, I examined the occurrences of the top-ranked verbs for advanced and beginner learner texts. In most cases, the strong attraction of particular verbs to the

[4]All instances of *have* are lexical, since auxiliary uses of *have* would have classified the construction as a third or mixed conditional.

verbal slot in the protasis of CIIs was driven by a high concentration of tokens in responses to a single writing task and within highly repetitive contexts. For instance, *face* appeared a total of 24 times in the whole dataset and only in texts responding to two prompts; notably, 22 of these occurrences were concentrated in responses to one single task, most often in repetitive contexts (almost all sentences contain the phrase *I knew that I would always regret it if I didn't face my fears*, with small variations). The exception to this pattern were the verbs *be* and *have*, which appear in responses to 15 and 11 tasks, respectively. Closer inspection revealed that even these highly dispersed verbs were strongly shaped by task-specific frames.

The verb *have* was particularly concentrated in responses to task 59 (*Making a "to do" list of your dreams*), where it occurred approximately five times more often than would be expected under a uniform distribution. All instances described hypotheticals about what the author would do if their circumstances were different. The if-clauses present considerable variation in their direct objects: *If I had more time*, *If I had a few millions*, or *If I had the opportunity*. Similarly, the distribution of *be* was heavily influenced by the prompts. In task 109 (*Talking a friend out of a risky action*), the verb occurred approximately six times more frequently than expected, always in the formulaic frequence for advising *If I were you*. The same pattern appeared in task 108 (*Attending a seminar on stress reduction*), where *be* was around three times more frequent than expected and likewise predominantly realized as *If I were you*. In task 74 (*Doing a survey about discrimination*), *be* occurred roughly four times more often than expected and in relatively more varied contexts, though always within the frame of questions directing the reader to imagine themselves in a hypothetical position: *If you were a man*, *If you were an HR director*, or *If you were a victim of discrimination*.

These findings suggest that many of the verbs identified as distinctive collexemes are closely tied to task-specific formulae and instructional prompts rather than reflecting creative use of the construction. Nevertheless, although seemingly shaped by the broader pedagogical context of the units, some variation in adjacent slots shows that the repetitive contexts do not hint to the construction not having been acquired productively.

A final consideration regarding DCA is that it is not possible to draw conclusions about the statistical significance of the differences between residuals based on this study, although one could assume the top-ranked verbs for either level to represent fairly robust associations. In order to ascertain whether the association strengths are higher than it would be expected by chance, an approach using bootstrapping like that outlined by Gries (2023) would be in order.

With regard to lexical diversity in the slot under discussion, one possible explanation for the similar entropy values across proficiency levels lies in the strong influence of task effects on verb choice. As shown in the DCA analysis, many verbs were highly concentrated in responses to particular prompts and frequently occurred in repetitive contexts. Such task-specific patterns may limit the extent to which increases in learners' available vocabulary are reflected in the overall distributional structure of the verbal slot.

Additionally, the slightly higher raw entropy observed for advanced learners should be interpreted cautiously. Because there is a larger number of tasks belonging to advanced levels and there are thus more instances of CIIs for this proficiency group overall, part of the increase may simply reflect greater opportunity for lexical variation rather than a broader range of verbs used in the construction. The comparatible normalized entropy values suggest that the overall patterns of verb use remain broadly similar across proficiency groups.

A potential question for future research is whether native-speaker production would exhibit substantially different entropy values. While native speakers would almost certainly display a wider range of collexemes, it is not necessarily clear that this would translate into higher normalized entropy.

5 Conclusion

This study compared the use of verbs in the if-clause of CIIs in writings by beginner (A2-B1) and advanced (B2-C1) learners of L2 English obtained from a CEFR-aligned, task-based corpus. It was shown that lower-proficiency learners produce the construction in similar proportions to their higher proficiency counterparts, though the task prompt and general educational context played a key role in its elicitation.

By means of DCA and subsequent manual inspection, lexical choice in the analyzed slot was revealed to be heavily influenced by the task prompt, with some collexemes being observed disproportionately often in specific tasks and within almost invariable contexts. Therefore, with the data and methods used in the present study, it was not possible to fully disentangle natural L2 development from task effects. In order to do this, a non-prompt based (or at least non-instructional) L2 corpus would be highly useful, although it is clear that the compilation of such a corpus would bring with it its own set of difficulties.

Finally, lexical diversity in the protasis verbal slot was operationalized by means of normalized entropy values, displaying comparable results between beginner and advanced texts. It is hypothesized that also this result is affected by task-specific effects together with the specific way of partitioning of the texts into the two proficiency levels. To mitigate these issues, an interesting future experiment might be to use a Large Language Model to generate approximations to native responses to the task prompts, enabling researchers to observe whether there is a substantial change in word choice and entropy values when native speakers are similarly constrained by tasks.

Despite these limitations, the present study demonstrates the usefulness of collostructional approaches for analyzing learner corpora at a fine-grained lexical level. By combining DCA with dispersion and lexical diversity measures, it is possible to move beyond purely frequency-based descriptions and examine how learners' lexical behavior within schematic constructions. More generally, the findings highlight the importance of considering context and task design when interpreting apparent developmental patterns in learner corpora, especially in educational settings where language production is influenced by instructional practices.

Appendices

A Task topics selected for analysis

Table 3 lists all the task topics whose responses were analyzed, along with their corresponding CEFR and *Englishtown* levels.

CEFR level	EF level:Unit	Topic ID	Words	Topic
A2	6:6	46	290,594	Writing an email of advice
B1	8:3	59	474,441	Making a ‘to do’ list of your dreams
B1	9:3	67	354,123	Making a business proposal
B1	9:4	68	284,752	Signing a waiver to go skydiving
B2	10:1	73	797,218	Helping a friend find a job
B2	10:2	74	492,941	Doing a survey about discrimination
B2	10:3	75	438,282	Requesting a bank loan
B2	10:5	77	313,135	Finding a home for a wealthy client
B2	11:2	82	291,016	Helping a coworker deal with a phobia
B2	11:8	88	24,119	Improving your study skills
C1	13:1	97	244,337	Writing a campaign speech
C1	13:4	100	115,523	Giving advice about budgeting
C1	13:8	104	21,223	Reaching your potential
C1	14:2	106	41,642	Choosing a renewable energy source
C1	14:3	107	61,155	Writing a rejection letter
C1	14:4	108	55,637	Attending a seminar on stress reduction
C1	14:5	109	52,968	Talking a friend out of a risky action
C1	14:6	110	39,465	Applying for sponsorship
C1	15:7	119	5,182	Writing about future lifestyles
C1	15:8	120	4,899	Interpreting a prophecy

Table 3: Topics by Level and CEFR

References

Theodora Alexopoulou, Marije Michel, Akira Murakami, and Detmar Meurers. Task effects on linguistic complexity and accuracy: A large-scale learner corpus analysis employing natural language processing techniques. *Language Learning*, 67(S1):180–208, 2017. doi: https://doi.org/10.1111/lang.12232. URL `https://onlinelibrary.wiley.com/doi/abs/10.1111/lang.12232`.

Annette Capel. A1–b2 vocabulary: insights and issues arising from the english profile wordlists project. *English Profile Journal*, 1:e3, 2010.

Annette Capel. Completing the english vocabulary profile: C1 and c2 vocabulary. *English Profile Journal*, 3:e1, 2012.

Nick C. Ellis. Formulaic language and second language acquisition: Zipf and the phrasal teddy bear. *Annual Review of Applied Linguistics*, 32:17–44, 2012. doi: 10.1017/S0267190512000025.

Jeroen Geertzen, Theodora Alexopoulou, and Anna Korhonen. Automatic linguistic annotation of large scale L2 databases: The EF-Cambridge Open Language Database (EFCAMDAT). In *Proceedings of the 31st Second Language Research Forum. Somerville, MA: Cascadilla Proceedings Project*, pages 240–254, 2013.

Stefan Th. Gries. Dispersions and adjusted frequencies in corpora. *International Journal of Corpus Linguistics*, 13(4):403–437, December 2008. ISSN 1384-6655, 1569-9811. doi: 10.1075/ijcl.13.4.02gri.

Stefan Th. Gries. Overhauling Collostructional Analysis: Towards More Descriptive Simplicity and More Explanatory Adequacy. *Cognitive Semantics*, 9(3):351–386, August 2023. ISSN 2352-6408, 2352-6416. doi: 10.1163/23526416-bja10056.

Stefan Th. Gries and Nick C. Ellis. Statistical Measures for Usage-Based Linguistics. *Language Learning*, 65(S1):228–255, 2015. ISSN 1467-9922. doi: 10.1111/lang.12119.

Stefan Th. Gries and Anatol Stefanowitsch. Extending collostructional analysis: A corpus-based perspective on 'alternations'. *International Journal of Corpus Linguistics*, 9(1):97–129, April 2004. ISSN 1384-6655, 1569-9811. doi: 10.1075/ijcl.9.1.06gri.

Stefan Th. Gries and Stefanie Wulff. Examining individual variation in learner production data: A few programmatic pointers for corpus-based analyses using the example of adverbial clause ordering. *Applied Psycholinguistics*, 42(2):279–299, 2021. doi: 10.1017/S014271642000048X.

Jefrey Lijffijt and Stefan Th. Gries. Correction to Stefan Th. Gries' "Dispersions and adjusted frequencies in corpora". *International Journal of Corpus Linguistics*, 17(1):147–149, 2012.

Anne O'Keeffe and Geraldine Mark. The english grammar profile of learner competence: Methodology and key findings. *International Journal of Corpus Linguistics*, 22(4):457–489, 2017.

Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D Manning. Stanza: A Python natural language processing toolkit for many human languages. In *Proceedings of the 58th annual meeting of the association for computational linguistics: system demonstrations*, pages 101–108, 2020.

Ute Römer-Barron. Learner corpus research in applied linguistics. *TESOL Quarterly*, 60(2):887–902, 2026. doi: https://doi.org/10.1002/tesq.70140. URL `https://onlinelibrary.wiley.com/doi/abs/10.1002/tesq.70140`.

Anatol Stefanowitsch and Stefan Th. Gries. Collostructions: Investigating the interaction of words and constructions. *International Journal of Corpus Linguistics*, 8(2):209–243, December 2003. ISSN 1384-6655, 1569-9811. doi: 10.1075/ijcl.8.2.03ste.

Kun Sun and Rong Wang. Using the relative entropy of linguistic complexity to assess l2 language proficiency development. *Entropy*, 23(8), 2021. ISSN 1099-4300. doi: 10.3390/e23081080. URL `https://www.mdpi.com/1099-4300/23/8/1080`.

Doğuş Öksüz, Theodora Alexopoulou, Kateryna Derkach, and Ianthi Maria Tsimpli. The influence of L1 typology on the acquisition of the L2 English article: A large-scale corpus study. *Second Language Research*, 2025. doi: 10.17863/CAM.122590. URL `https://www.repository.cam.ac.uk/handle/1810/391460`.